Skills Summary

Solving AAS and ASA Triangles with the Law of Sines

Use this reference while completing your worksheet or when studying.

Skill 1 — Naming the Case and Finding the Third Angle

Angle sum: $A + B + C = 180°$. **Naming rule:** side a is opposite angle A, and so on.

- **AAS**: the given side is opposite one of the given angles $\Rightarrow$ a complete side–angle pair already exists.
- **ASA**: the given side lies *between* the given angles $\Rightarrow$ no pair yet, so find the third angle first.
- Either way, two angles force the third, so the triangle's shape is fixed and one side sets its size.

Example: $A = 37°$, $B = 58°$, $a = 10$. Find C and name the case.
Step 1. Two angles are known, so the third is forced by the angle sum — no trigonometry is needed for this step.
Step 2. $C = 180° - 37° - 58° = \mathbf{85°}$. Side a is opposite the given A, so this is AAS.

Try it: $A = 64°$, $C = 51°$, $b = 10$. Find B.

check: $\boldsymbol{B = 65°}$

Skill 2 — AAS: Use the Complete Pair Straight Away

Law of sines: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

To find an unknown **side**, put the unknown on top:

$$\text{unknown} = \frac{(\text{known side})\ \sin(\text{angle opposite the unknown})}{\sin(\text{angle opposite the known side})}$$

Sanity check: the bigger angle must always face the longer side.

Example: $A = 41°$, $B = 79°$, $a = 14$. Find b.
Step 1. Side a is opposite the given angle A, so the pair $a/\sin A$ is complete and the law of sines can be used at once.
Step 2. $b = \dfrac{14 \sin 79°}{\sin 41°} = \dfrac{14(0.981627)}{0.656059} = 20.9475\ldots \Rightarrow \boldsymbol{b \approx 20.95}$, and $B > A$ so $b > a$ as expected.

Try it: $A = 36°$, $B = 68°$, $a = 20$. Find b.

check: $\boldsymbol{b \approx 31.55}$

Skill 3 — ASA: Third Angle First, Then the Law of Sines

Why the extra step: in ASA the known side sits *between* the known angles, so the angle opposite it is the one you do not have. Build the pair first:

- Step 1: third angle from 180° minus the two given angles.
- Step 2: the known side is now paired with that new angle — apply the law of sines.
- Keep full calculator precision and round only the final side.

Example: $A = 54°$, $B = 39°$, $c = 25$. Find a.
Step 1. Side c lies between the two given angles, so no pair is complete — the angle opposite c must be found before the law of sines can be used.
Step 2. $C = 180° - 54° - 39° = 87°$, then $a = \dfrac{25 \sin 54°}{\sin 87°} = 20.2532\ldots \Rightarrow \boldsymbol{a \approx 20.25}$

Try it: $A = 47°$, $B = 62°$, $c = 30$. Find a.

check: $\boldsymbol{a \approx 23.20}$

(!) Watch out: Match each side with the angle *opposite* it, never with the nearest one — an upside-down ratio is the single most common error here. And in ASA, skipping the third angle leaves you with no complete pair at all.